Lecture-code

```r
# x <- scan() # wont accept string
floor(6.3) # same as 6.3%/%1
```

```
[1] 6
```

```r
temp <- c(273, 283, 293, 303, 313, 323, 333, 343, 353, 363, 373)
length(temp)
```

```
[1] 11
```

```r
r_temp <- rev(temp)
summary(r_temp)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
    273     298     323     323     348     373 
```

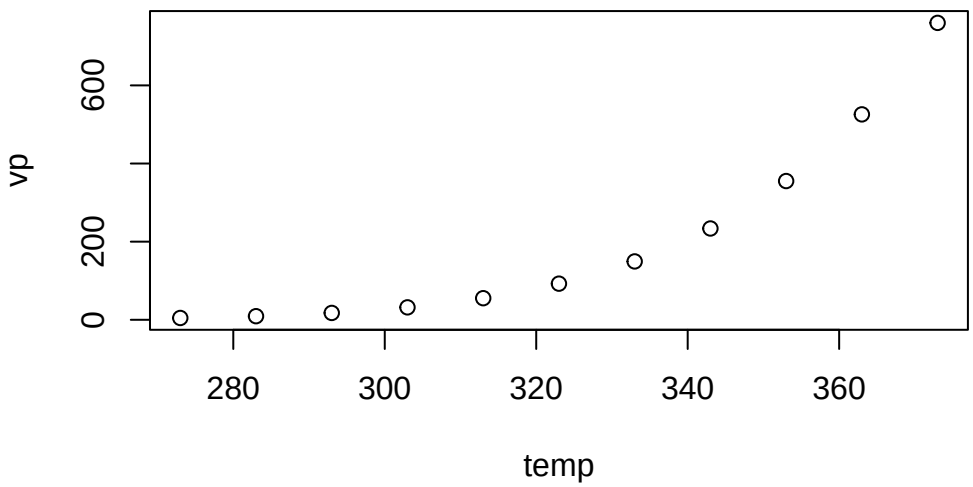
```r
vp <- c(4.6, 9.2, 17.5, 31.8, 55.3, 92.5, 149.4, 233.7, 355.1, 525.8, 760.0)
vapor <- data.frame(temp, vp)
vapor$temp
```

```
 [1] 273 283 293 303 313 323 333 343 353 363 373
```

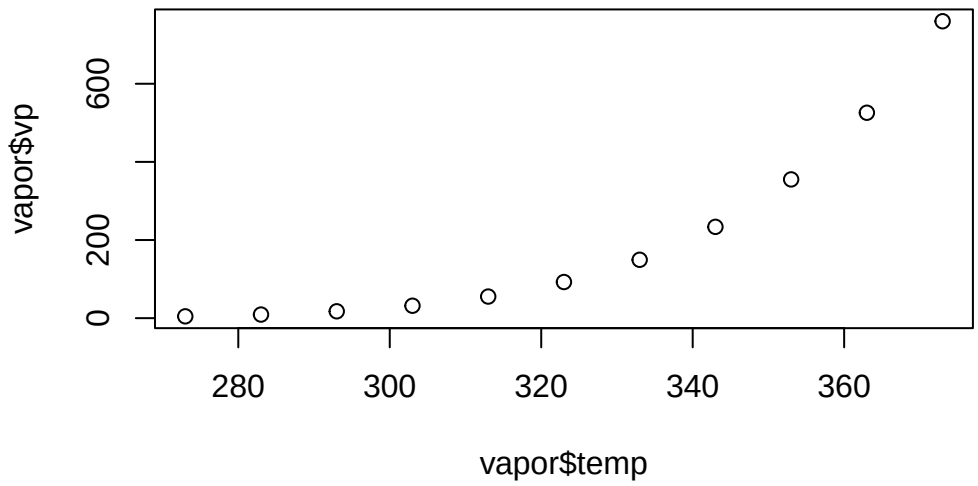
```r
vapor$vp
```

```
 [1]   4.6   9.2  17.5  31.8  55.3  92.5 149.4 233.7 355.1 525.8 760.0
```

```r
inv_temp <- 1/vapor$temp
plot(vapor)
```




```r
plot(vapor$temp, vapor$vp)
```




```r
# generating numbers
seq(0,1,length = 15)
```

```
 [1] 0.00000000 0.07142857 0.14285714 0.21428571 0.28571429 0.35714286
 [7] 0.42857143 0.50000000 0.57142857 0.64285714 0.71428571 0.78571429
[13] 0.85714286 0.92857143 1.00000000
```

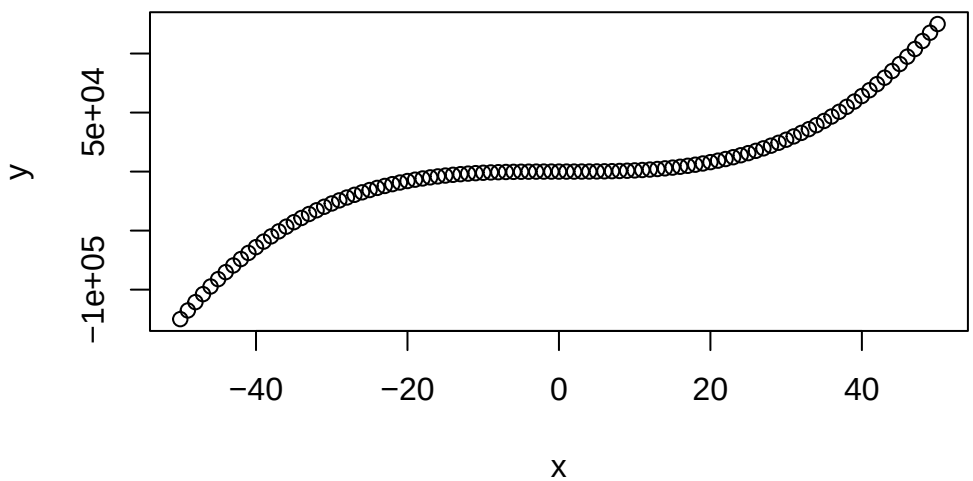
```r
rep("A", 10)
```

```
 [1] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
```

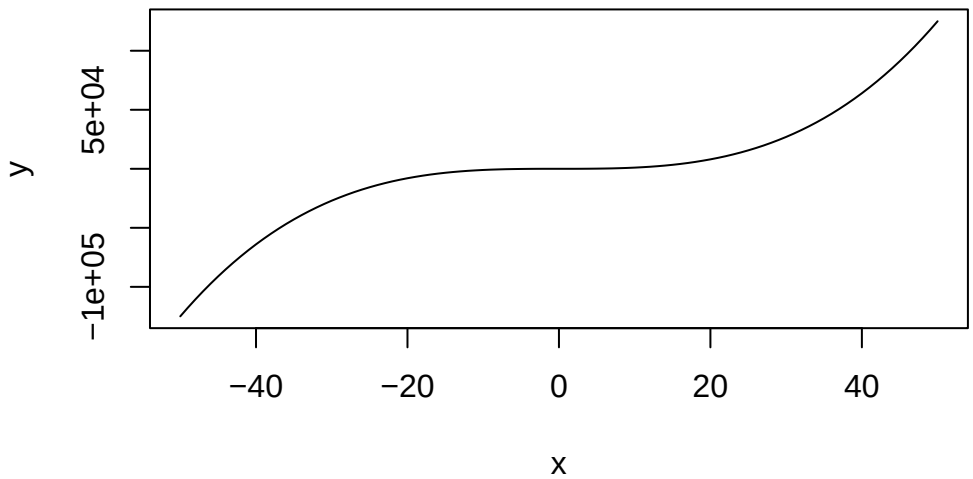
```r
runif(50, 0, 1)
```

```
 [1] 0.53000739 0.62522522 0.23538461 0.34553524 0.28291961 0.20168674
 [7] 0.11982252 0.40214854 0.42338864 0.05366505 0.52723391 0.76726484
[13] 0.51614113 0.86347846 0.06912418 0.05699780 0.29178055 0.20416449
[19] 0.76771710 0.31000833 0.91908545 0.12942688 0.62293963 0.43614824
[25] 0.31364942 0.14931789 0.61207239 0.20035407 0.41506892 0.84401784
[31] 0.43888983 0.77572627 0.13015889 0.16848561 0.80795309 0.80352386
[37] 0.38185398 0.92938720 0.31837687 0.02473528 0.04613130 0.76126447
[43] 0.13333946 0.24455750 0.32346051 0.67142561 0.35632094 0.95259255
[49] 0.49079676 0.64308194
```

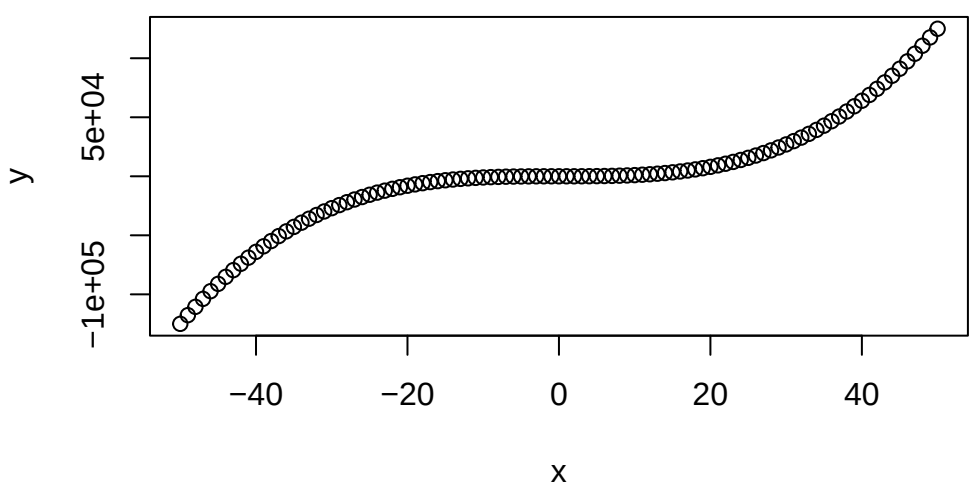
```r
# plotting
x <- -50:50
y <-  x^3

plot(x,y, type = "p")
```



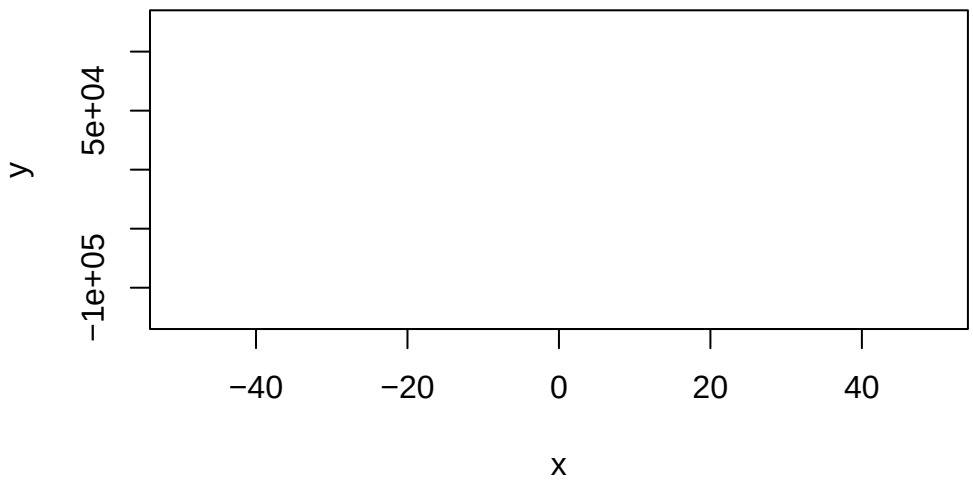

```r
plot(x,y, type = "l")
```



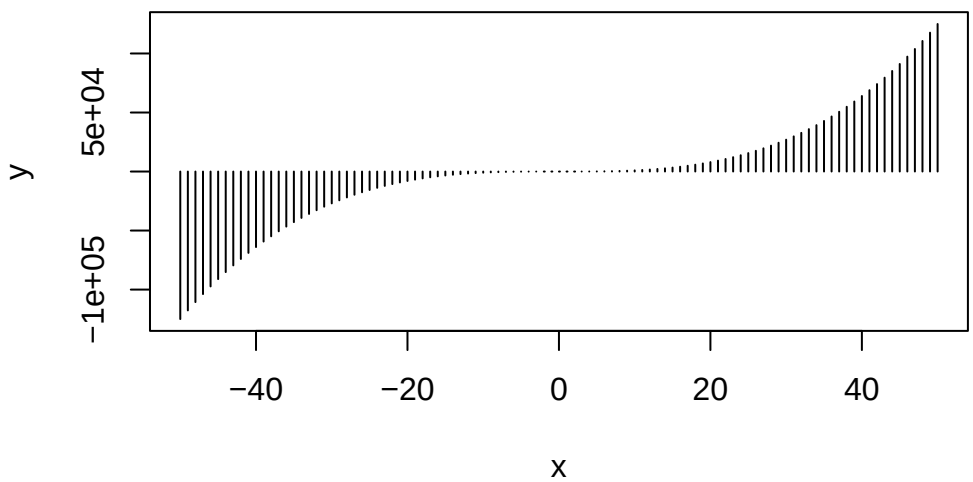

```r
plot(x,y, type = "b")
```



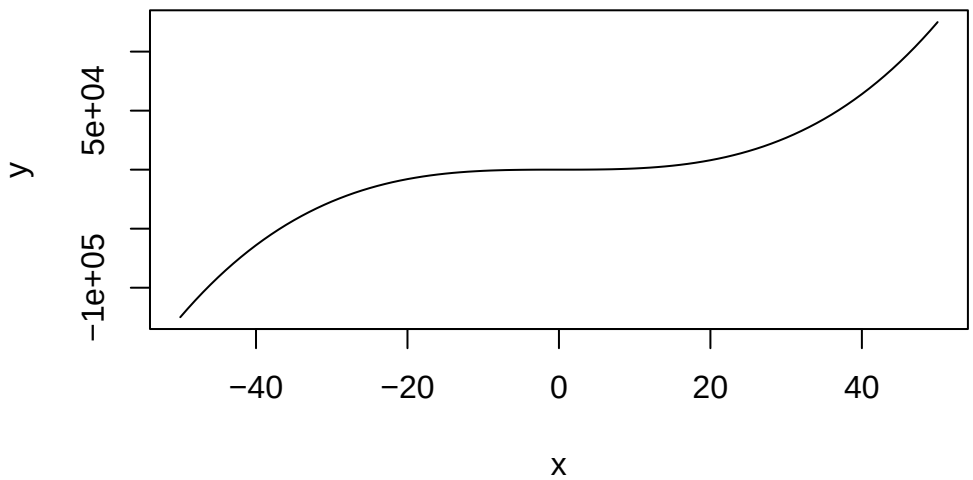
```r
plot(x,y, type = "n")
```



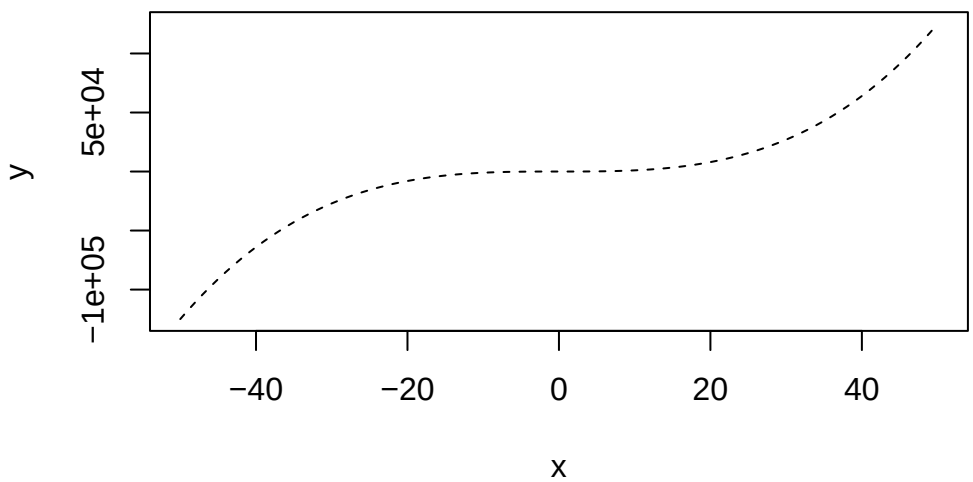

```r
plot(x,y, type = "h")
```



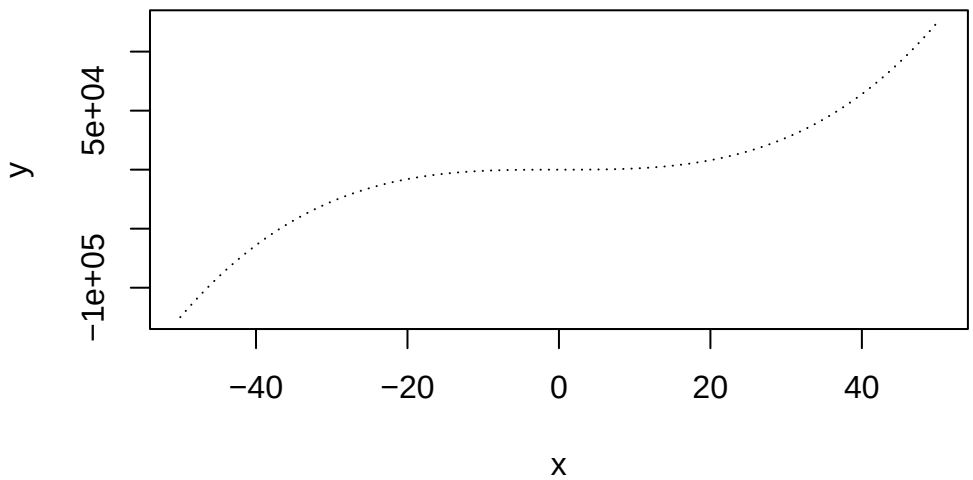
```r
plot(x, y, type = "l", lty = 1)
```



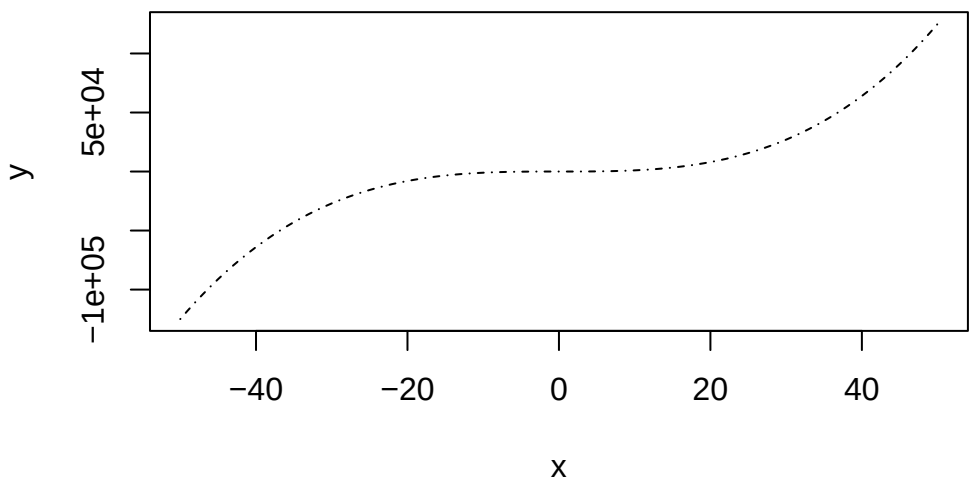

```r
plot(x, y, type = "l", lty = 2)
```



```r
plot(x, y, type = "l", lty = 3)
```




```r
plot(x, y, type = "l", lty = 4)
```

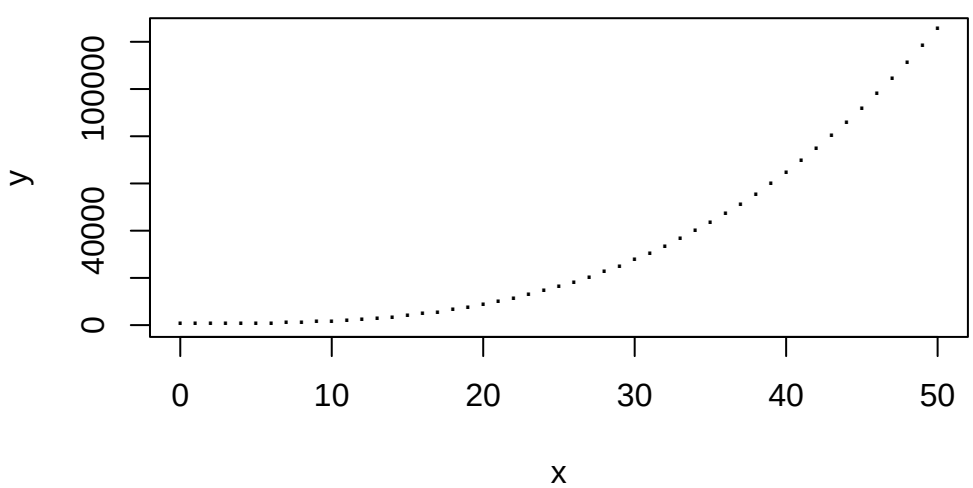


```r
x <- 0:50
y <-  x^3
plot(x, y, type = "p", pch = " ")
```

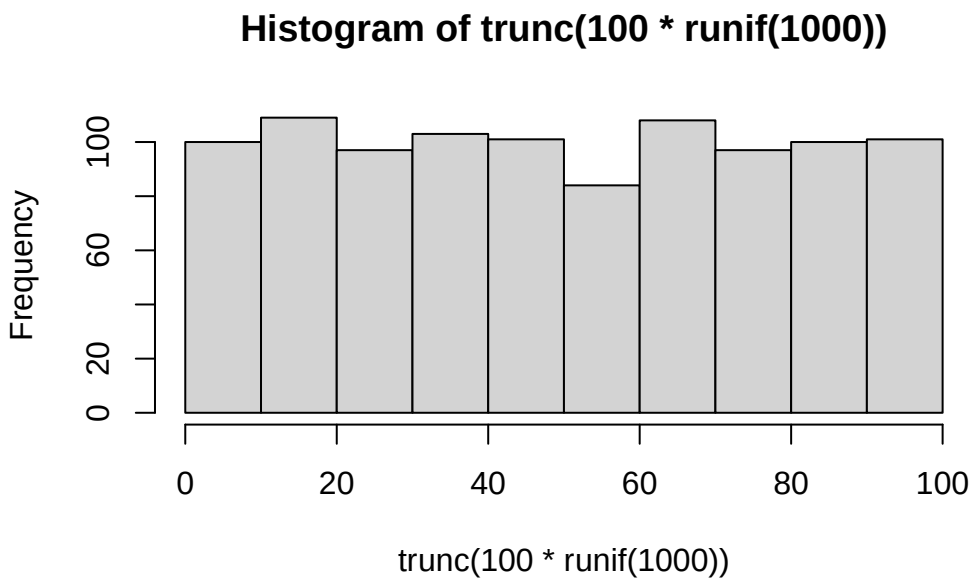
```
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
```

```
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
```

```
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
Warning in plot.xy(xy, type, ...): font metrics unknown for Unicode character
U+1F480
```

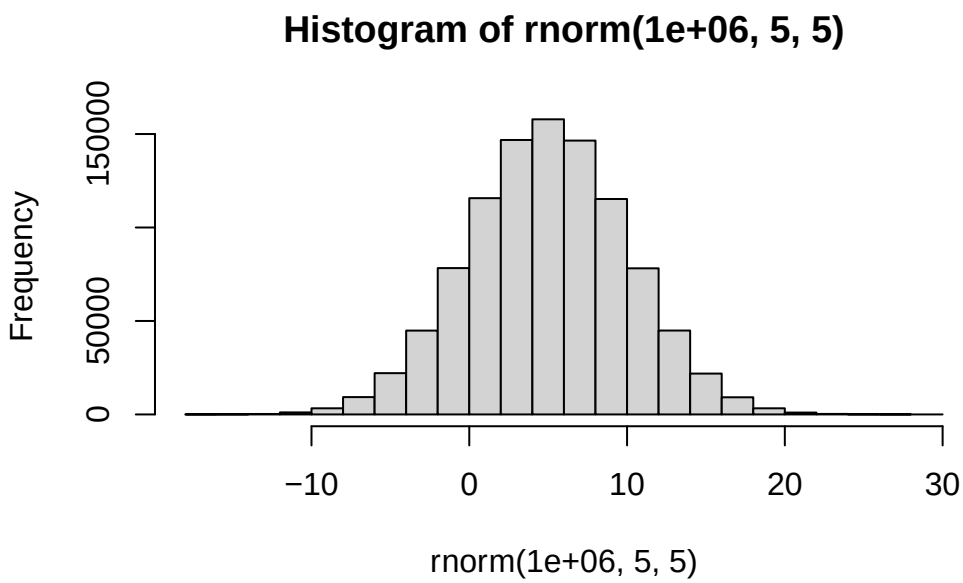


```r
(hist(trunc(100*runif(1000))))
```




```
$breaks
 [1]   0  10  20  30  40  50  60  70  80  90 100

$counts
 [1] 100 109  97 103 101  84 108  97 100 101

$density
 [1] 0.0100 0.0109 0.0097 0.0103 0.0101 0.0084 0.0108 0.0097 0.0100 0.0101

$mids
 [1]  5 15 25 35 45 55 65 75 85 95

$xname
[1] "trunc(100 * runif(1000))"

$equidist
[1] TRUE

attr(,"class")
[1] "histogram"
```

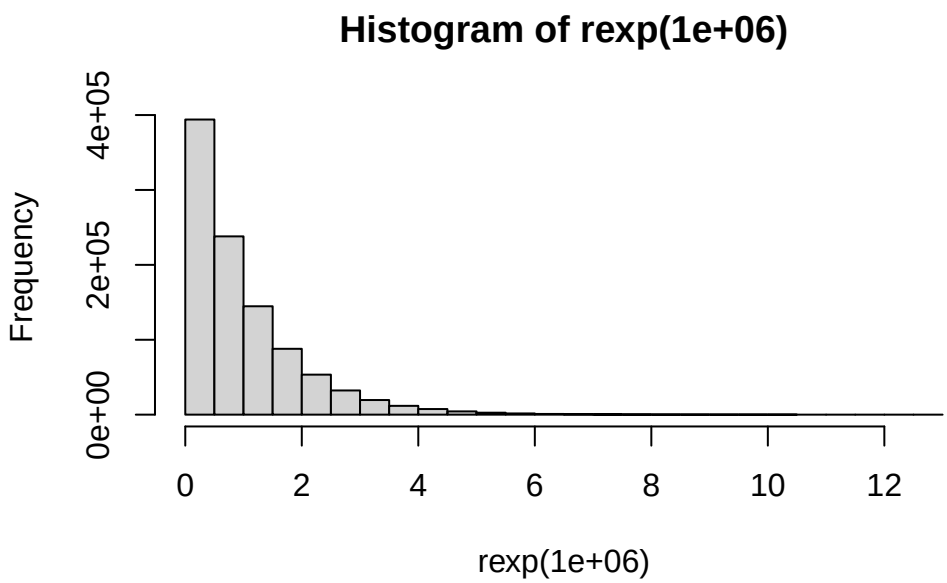
```r
(hist(rnorm(1e6,5,5)))
```




```
$breaks
 [1] -18 -16 -14 -12 -10  -8  -6  -4  -2   0   2   4   6   8  10  12  14  16  18
[20]  20  22  24  26  28  30

$counts
 [1]      11      62     239    1087    3283    9269   22083   44852   78292  115692
[11]  146819  157872  146495  115256   78163   44883   21856    9170    3277    1005
[21]     264      55      14       1

$density
 [1] 0.0000055 0.0000310 0.0001195 0.0005435 0.0016415 0.0046345 0.0110415
 [8] 0.0224260 0.0391460 0.0578460 0.0734095 0.0789360 0.0732475 0.0576280
[15] 0.0390815 0.0224415 0.0109280 0.0045850 0.0016385 0.0005025 0.0001320
[22] 0.0000275 0.0000070 0.0000005

$mids
 [1] -17 -15 -13 -11  -9  -7  -5  -3  -1   1   3   5   7   9  11  13  15  17  19
[20]  21  23  25  27  29

$xname
```

```
[1] "rnorm(1e+06, 5, 5)"

$equidist
[1] TRUE

attr(,"class")
[1] "histogram"
```

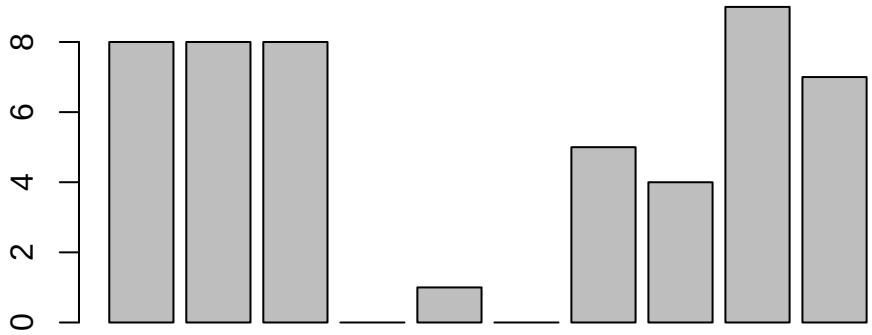
```r
(hist(rexp(1e6), nclass = 35))
```



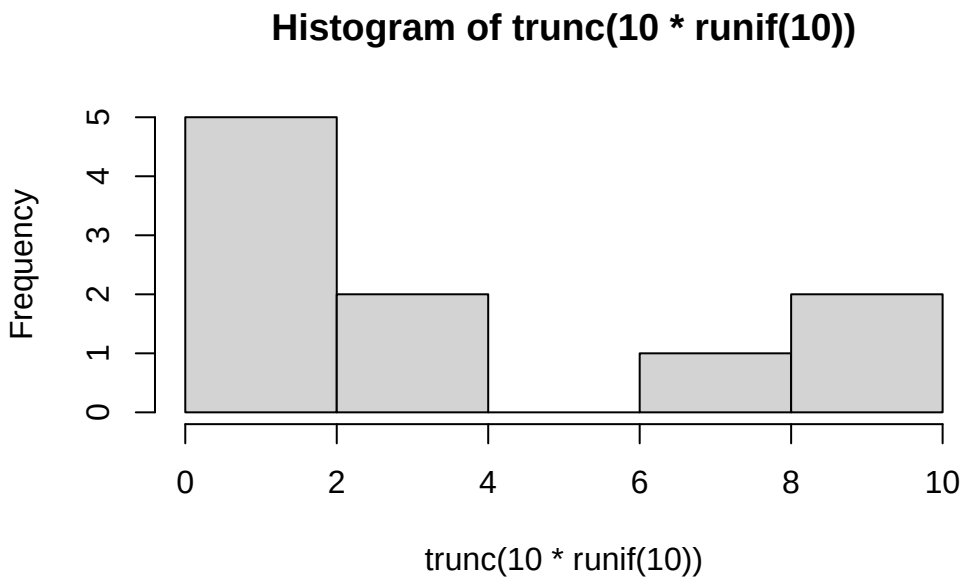

```
$breaks
 [1]  0.0  0.5  1.0  1.5  2.0  2.5  3.0  3.5  4.0  4.5  5.0  5.5  6.0  6.5  7.0
[16]  7.5  8.0  8.5  9.0  9.5 10.0 10.5 11.0 11.5 12.0 12.5 13.0

$counts
 [1] 394012 238018 144685  87908  53394  32292  19507  11765   7491   4326
[11]   2588   1592    939    592    383    228    116     72     38     18
[21]     20      3      5      5      2      1

$density
 [1] 0.788024 0.476036 0.289370 0.175816 0.106788 0.064584 0.039014 0.023530
 [9] 0.014982 0.008652 0.005176 0.003184 0.001878 0.001184 0.000766 0.000456
```

```
[17] 0.000232 0.000144 0.000076 0.000036 0.000040 0.000006 0.000010 0.000010
[25] 0.000004 0.000002

$mids
 [1]  0.25  0.75  1.25  1.75  2.25  2.75  3.25  3.75  4.25  4.75  5.25  5.75
[13]  6.25  6.75  7.25  7.75  8.25  8.75  9.25  9.75 10.25 10.75 11.25 11.75
[25] 12.25 12.75

$xname
[1] "rexp(1e+06)"

$equidist
[1] TRUE

attr(,"class")
[1] "histogram"
```

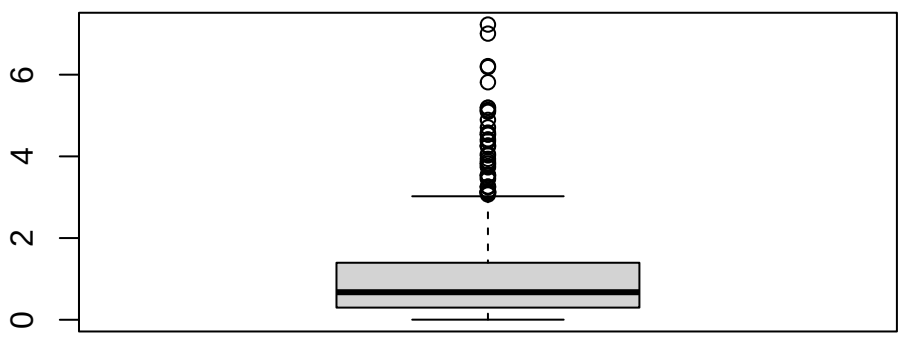
```r
barplot(trunc(10*runif(10)))
```



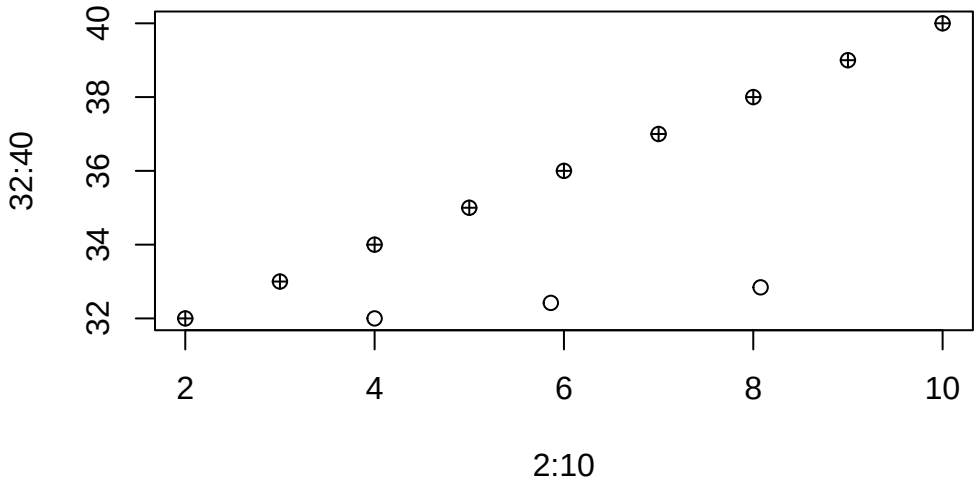

```r
hist(trunc(10*runif(10)))
```



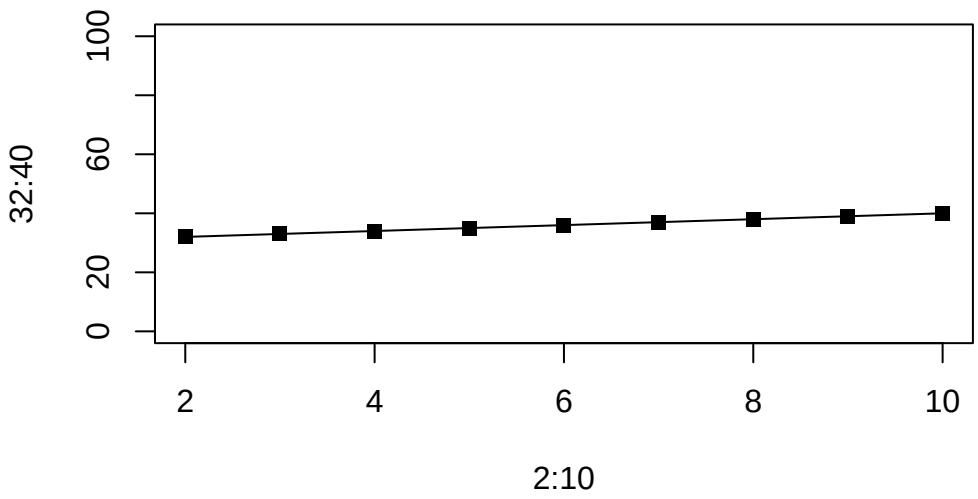

```r
boxplot(rexp(1000))
```



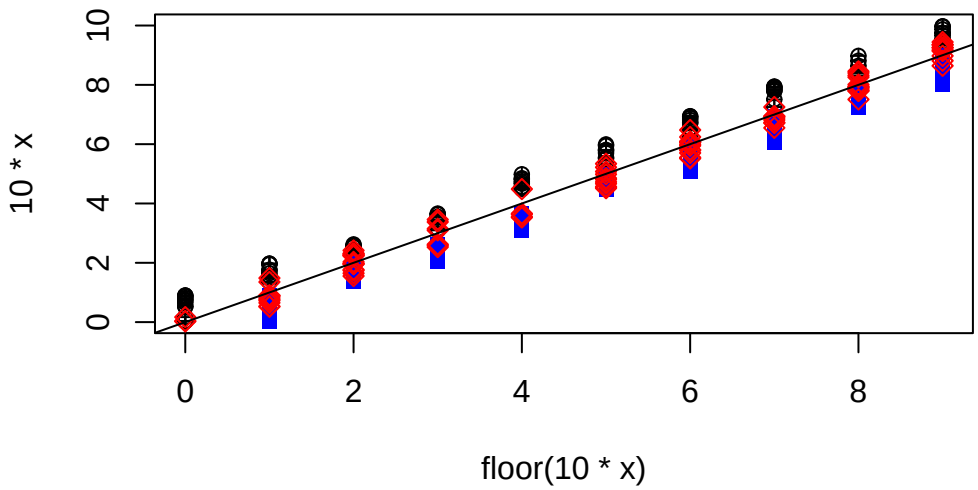
```r
plot(2:10, 32:40, pch = 10)
x <- seq(2, 10, length = 20)
y <- seq(32, 40, length = 20)
points(x^2, y)
```




```r
plot(2:10, 32:40, pch = 15, ylim = c(0,100))
lines(x, y)
```



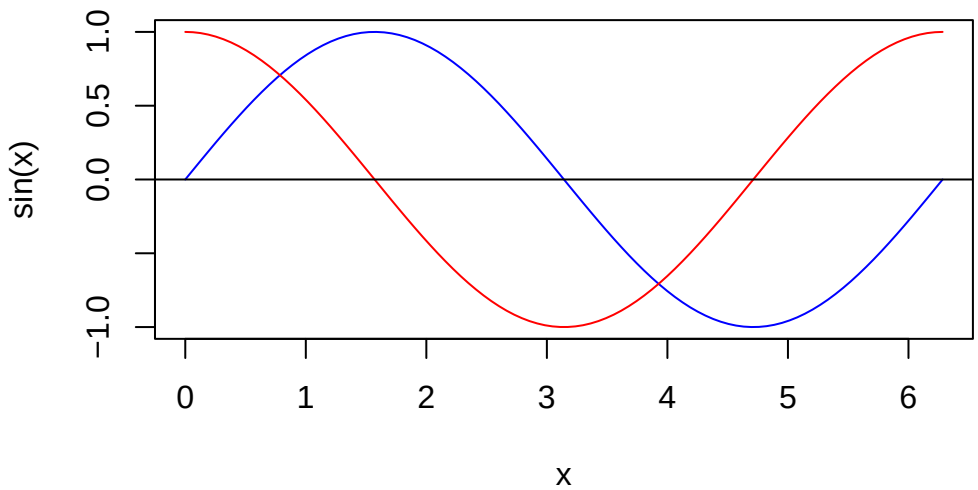

```r
x <- runif(100)

plot(floor(10*x), 10*x, pch = 10)
points(ceiling(10*x), 10*x, pch = 15, col = "blue")
points(round(10*x), 10*x, pch = 5, col = "red")
abline(0,1)
```




```r
x <- seq(0, 2*pi, length = 100)
x
```

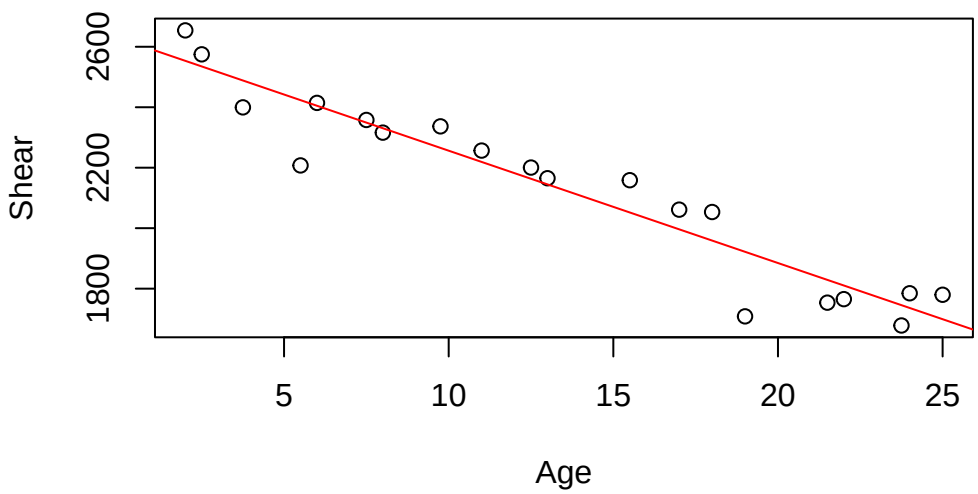
```
  [1] 0.00000000 0.06346652 0.12693304 0.19039955 0.25386607 0.31733259
  [7] 0.38079911 0.44426563 0.50773215 0.57119866 0.63466518 0.69813170
 [13] 0.76159822 0.82506474 0.88853126 0.95199777 1.01546429 1.07893081
 [19] 1.14239733 1.20586385 1.26933037 1.33279688 1.39626340 1.45972992
 [25] 1.52319644 1.58666296 1.65012947 1.71359599 1.77706251 1.84052903
 [31] 1.90399555 1.96746207 2.03092858 2.09439510 2.15786162 2.22132814
 [37] 2.28479466 2.34826118 2.41172769 2.47519421 2.53866073 2.60212725
 [43] 2.66559377 2.72906028 2.79252680 2.85599332 2.91945984 2.98292636
 [49] 3.04639288 3.10985939 3.17332591 3.23679243 3.30025895 3.36372547
 [55] 3.42719199 3.49065850 3.55412502 3.61759154 3.68105806 3.74452458
 [61] 3.80799110 3.87145761 3.93492413 3.99839065 4.06185717 4.12532369
 [67] 4.18879020 4.25225672 4.31572324 4.37918976 4.44265628 4.50612280
 [73] 4.56958931 4.63305583 4.69652235 4.75998887 4.82345539 4.88692191
 [79] 4.95038842 5.01385494 5.07732146 5.14078798 5.20425450 5.26772102
 [85] 5.33118753 5.39465405 5.45812057 5.52158709 5.58505361 5.64852012
 [91] 5.71198664 5.77545316 5.83891968 5.90238620 5.96585272 6.02931923
 [97] 6.09278575 6.15625227 6.21971879 6.28318531
```

```r
plot(x, sin(x), type = "l", col = "blue")
points(x, cos(x), type = "l", col = "red")
abline(h=0)
```




```r
# R code Example 2.1

Age<-c(15.50, 23.75, 8.00, 17.00, 5.50, 19.00, 24.00, 2.50, 7.50, 11.00, 13.00, 3.75, 25.00,
Shear<-c(2158.70, 1678.15, 2316.00, 2061.30, 2207.50, 1708.30, 1784.70, 2575.00, 2357.90, 225

plot(Age, Shear)  # Scatter Plot


Rocket.Data<-data.frame(Shear, Age)  # Define the data frame

# Finding regression parameters
# First method using the formulas

n<-length(Age)
x<-Age ; y<-Shear
Sxx<-sum(x^2)-(sum(x))^2/n  # sum of squares of x
Sxy<-sum(x*y)-sum(x)*sum(y)/n   # sum of squares of xy
SST <- sum(y^2) - (sum(y))^2/n  # Total sum of squares
```

```r
SSRes <- SST - Sxy^2/Sxx                # Regression sum of squares
hatbeta1<-Sxy/Sxx
hatbeta0<-mean(y)-hatbeta1*mean(x)
hatsigma2<-SSRes/(n-2)  # estimate of variance of error term

errorbeta1<-sqrt(hatsigma2/Sxx)  # standard error of beta1
errorbeta0<-sqrt(hatsigma2*(1/n + (mean(x))^2/Sxx))  # standard error of beta0

abline(hatbeta0, hatbeta1, col = "red")
```




```r
# testing hypothesis beta1 = 0
(tvalue<-hatbeta1/errorbeta1)
```

```
[1] -12.85989
```

```r
# p-value for two tail test
(2*(1 - pt(abs(tvalue), df = n-2)))
```

```
[1] 1.643343e-10
```

```r
(1 - pt(abs(tvalue), df = n-2))
```

```
[1] 8.216716e-11
```

```r
qt(0.975, df = n-2)  # critical value for 95% confidence interval
```

```
[1] 2.100922
```

```r
qnorm(0.975)
```

```
[1] 1.959964
```

```r
# testing hypothesis beta0 = 0
(tvalue0<-hatbeta0/errorbeta0)
```

```
[1] 59.47464
```

```r
# p-value for two tail test
(2*(1 - pt(abs(tvalue0), df = n-2)))
```

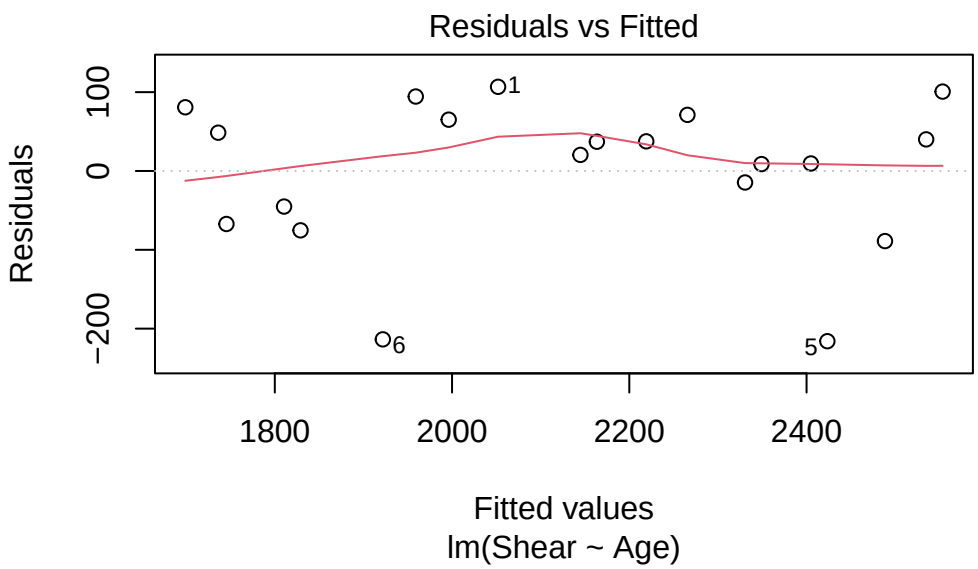
```
[1] 0
```

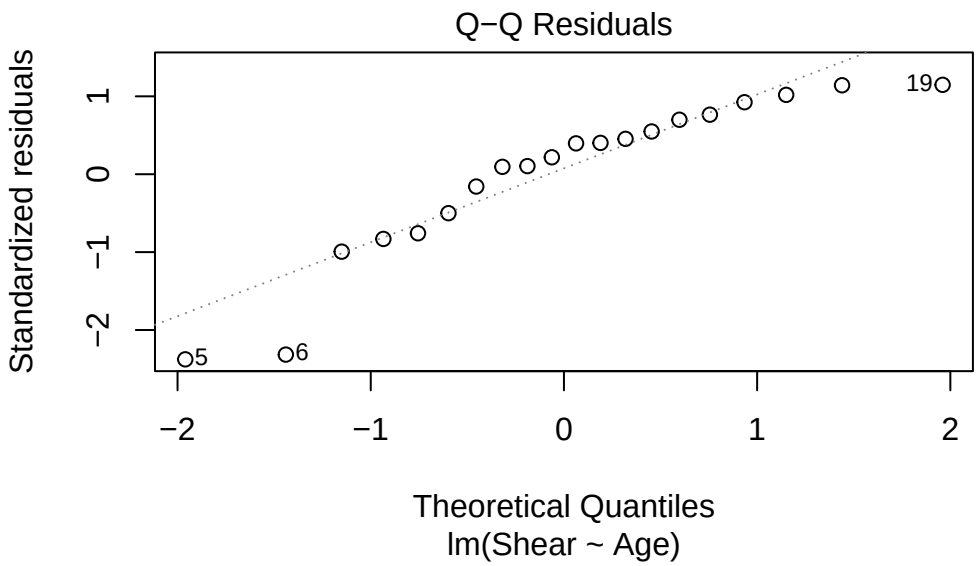
```r
# plotting tdistribution




#####################################
# Second Method  #  R lm Code

Out1<-lm(Shear~Age) # Get regression coefficients
summary(Out1)
```

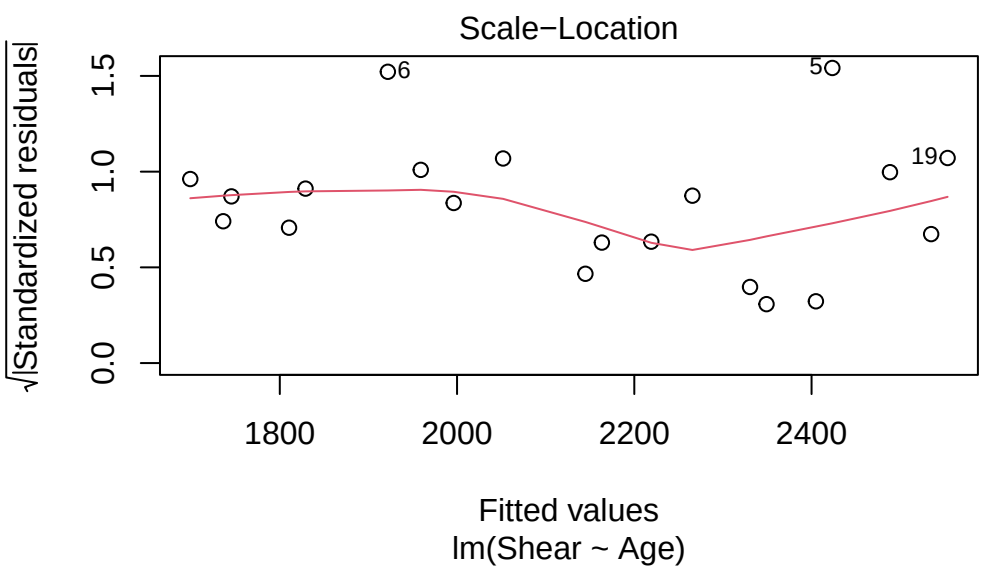
```
Call:
lm(formula = Shear ~ Age)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-215.98  -50.68   28.74   66.61  106.76 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 2627.822     44.184   59.48  < 2e-16 ***
Age          -37.154      2.889  -12.86 1.64e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 96.11 on 18 degrees of freedom
Multiple R-squared:  0.9018,	Adjusted R-squared:  0.8964 
F-statistic: 165.4 on 1 and 18 DF,  p-value: 1.643e-10
```

```r
plot(Out1)
```





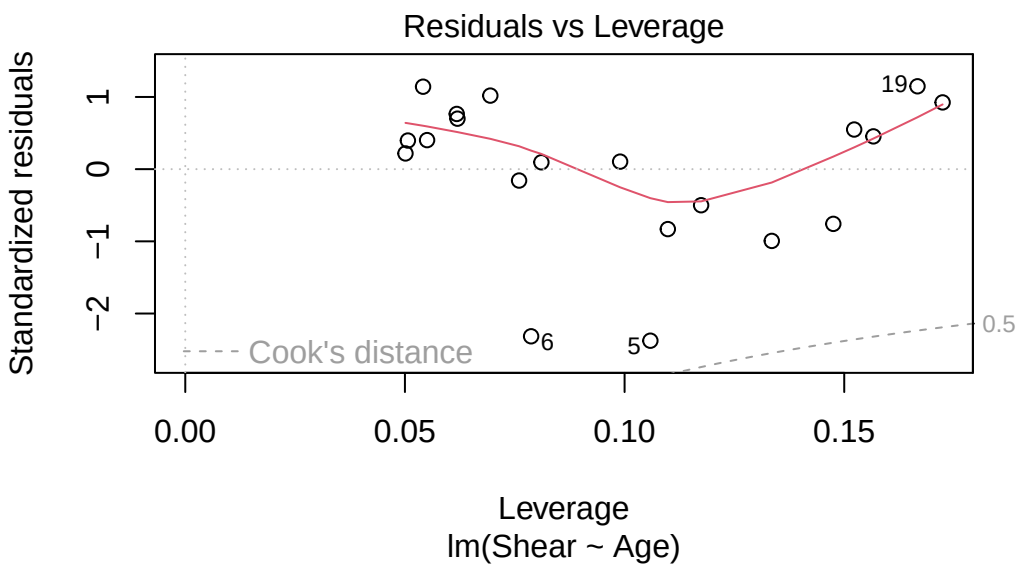



```
anova(Out1)
```

```
Analysis of Variance Table
```

```
Response: Shear
          Df  Sum Sq Mean Sq F value    Pr(>F)
Age        1 1527483 1527483  165.38 1.643e-10 ***
Residuals 18  166255    9236
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
abline(2627.82, -37.15)  # to add regression line
```




```r
# For more detailes Regression outputs

(fitted <- 2627.822 -37.154 *x)
```

```
 [1] 2051.935 1745.415 2330.590 1996.204 2423.475 1921.896 1736.126 2534.937
 [9] 2349.167 2219.128 2144.820 2488.495 1698.972 2265.571 1810.434 1959.050
[17] 2404.898 2163.397 2553.514 1829.011
```

```r
fitted(Out1)  # hat y
```

```
        1         2         3         4         5         6         7         8
```

```
2051.942 1745.425 2330.594 1996.211 2423.478 1921.904 1736.136 2534.938
       9       10       11       12       13       14       15       16
2349.170 2219.133 2144.826 2488.496 1698.983 2265.575 1810.443 1959.058
      17       18       19       20
2404.901 2163.402 2553.515 1829.020
```

```r
Out1$fitted.values
```

```
       1        2        3        4        5        6        7        8
2051.942 1745.425 2330.594 1996.211 2423.478 1921.904 1736.136 2534.938
       9       10       11       12       13       14       15       16
2349.170 2219.133 2144.826 2488.496 1698.983 2265.575 1810.443 1959.058
      17       18       19       20
2404.901 2163.402 2553.515 1829.020
```

```r
(redisuals <- y - fitted)
```

```
 [1]  106.7650  -67.2645  -14.5900   65.0960 -215.9750 -213.5960   48.5740
 [8]   40.0630    8.7330   37.5720   20.3800  -88.9445   80.8280   71.1795
[15]  -45.1340   94.4500    9.5020   37.1030  100.6860  -75.3110
```

```r
resid(Out1)   # residuals
```

```
          1           2           3           4           5           6
 106.758301  -67.274574  -14.593631   65.088687 -215.977609 -213.604131
          7           8           9          10          11          12
  48.563824   40.061618    8.729573   37.567141   20.374323  -88.946393
         13          14          15          16          17          18
  80.817415   71.175153  -45.143358   94.442278    9.499187   37.097528
         19          20
 100.684823  -75.320154
```

```r
Out1$residuals
```

```
          1           2           3           4           5           6
 106.758301  -67.274574  -14.593631   65.088687 -215.977609 -213.604131
          7           8           9          10          11          12
  48.563824   40.061618    8.729573   37.567141   20.374323  -88.946393
         13          14          15          16          17          18
  80.817415   71.175153  -45.143358   94.442278    9.499187   37.097528
         19          20
 100.684823  -75.320154
```

```r
anova(Out1)   # Analysis of Variance Table
```

```
Analysis of Variance Table

Response: Shear
          Df  Sum Sq Mean Sq F value    Pr(>F)
Age        1 1527483 1527483  165.38 1.643e-10 ***
Residuals 18  166255    9236
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# example 1
x <- c(24.00, 29.00, 24.00, 22.50, 17.75, 21.50, 24.00, 28.50, 21.00, 24.00, 23.50, 20.00)
y <- c(34.00, 33.69, 35.00,35.00, 28.00, 34.00, 29.50, 30.00, 28.00, 33.00, 17.50, 28.00)
plot(x, y)

grades.fit <- lm(x ~ y)
summary(grades.fit)
```

```
Call:
lm(formula = x ~ y)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2184 -2.0520  0.1276  1.1153  5.2535

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  19.0739     6.1395   3.107   0.0111 *
y             0.1391     0.1991   0.699   0.5007
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.274 on 10 degrees of freedom
Multiple R-squared:  0.04655,   Adjusted R-squared:  -0.0488
F-statistic: 0.4882 on 1 and 10 DF,  p-value: 0.5007
```

```r
n<-length(x)
Sxx<-sum(x^2)-(sum(x))^2/n  # sum of squares of x
```

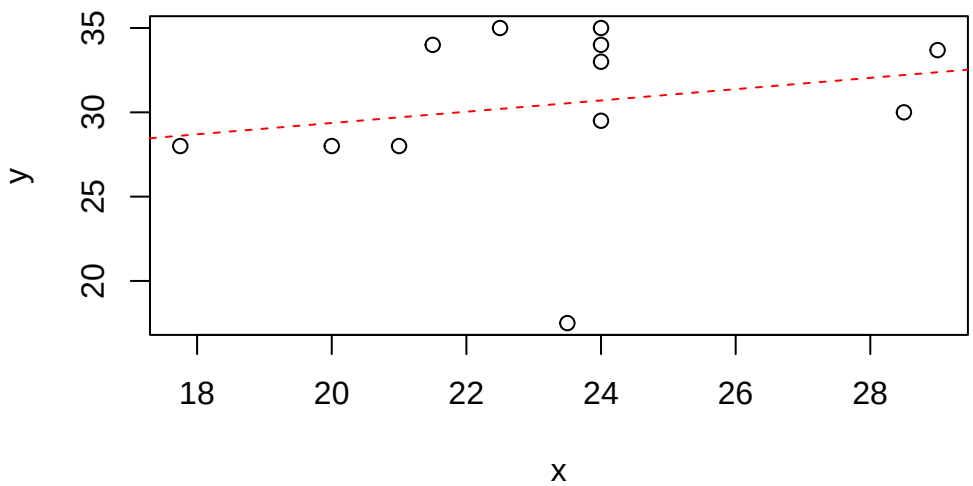
```r
Sxy<-sum(x*y)-sum(x)*sum(y)/n   # sum of squares of xy

(hatbeta1<-Sxy/Sxx)
```

```
[1] 0.3346531
```

```r
(hatbeta0<-mean(y)-hatbeta1*mean(x))
```

```
[1] 22.67257
```

```r
abline(hatbeta0, hatbeta1, col = "red", lty = "dashed")
```




```r
# example 2
x <- c(4, -1, 2, 3, 5, 4, 1)
y <- c(15, 10, 16, 12, 18, 17, 14)
plot(x, y, main = "Scatter plot: x vs y")

test.fit <- lm(x ~ y)
summary(grades.fit)
```

```
Call:
lm(formula = x ~ y)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.2184 -2.0520  0.1276  1.1153  5.2535 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)  19.0739     6.1395   3.107   0.0111 *
y             0.1391     0.1991   0.699   0.5007  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.274 on 10 degrees of freedom
Multiple R-squared:  0.04655,	Adjusted R-squared:  -0.0488 
F-statistic: 0.4882 on 1 and 10 DF,  p-value: 0.5007
```

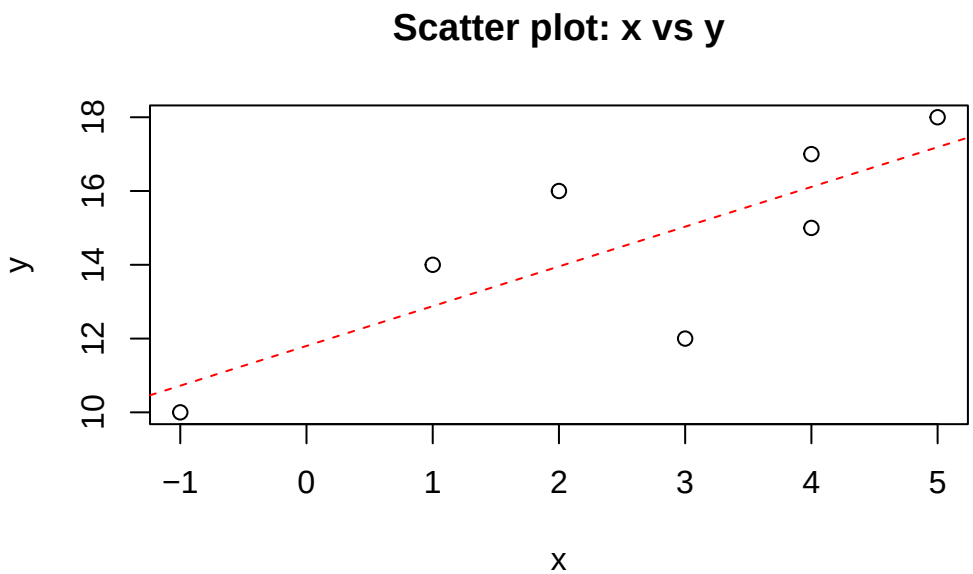
```r
n<-length(x)
Sxx<-sum(x^2)-(sum(x))^2/n  # sum of squares of x
Sxy<-sum(x*y)-sum(x)*sum(y)/n   # sum of squares of xy

(hatbeta1<-Sxy/Sxx)
```

```
[1] 1.077778
```

```r
(hatbeta0<-mean(y)-hatbeta1*mean(x))
```

```
[1] 11.8
```

```r
abline(hatbeta0, hatbeta1, col = "red", lty = "dashed")
```




```r
(mean(test.fit$residuals))
```

```
[1] -6.344132e-17
```

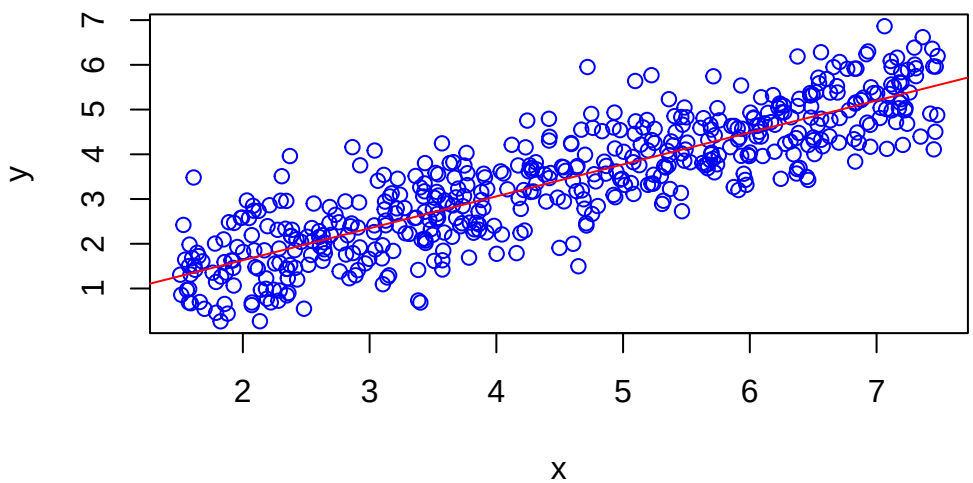
```r
test.fit$fitted.values
```

```
          1           2           3           4           5           6 
 2.82035928 -0.08383234  3.40119760  1.07784431  4.56287425  3.98203593 
          7 
 2.23952096 
```

```r
income_data <- read.csv("incomedata.csv")
head(income_data)
```

```
  X   income happiness
1 1 3.862647  2.314489
2 2 4.979381  3.433490
3 3 4.923957  4.599373
4 4 3.214372  2.791114
5 5 7.196409  5.596398
6 6 3.729643  2.458556
```

```r
x <- income_data$income
y <- income_data$happiness
plot(x, y, col = "blue")
abline(lm(y ~ x), col = "red")
```

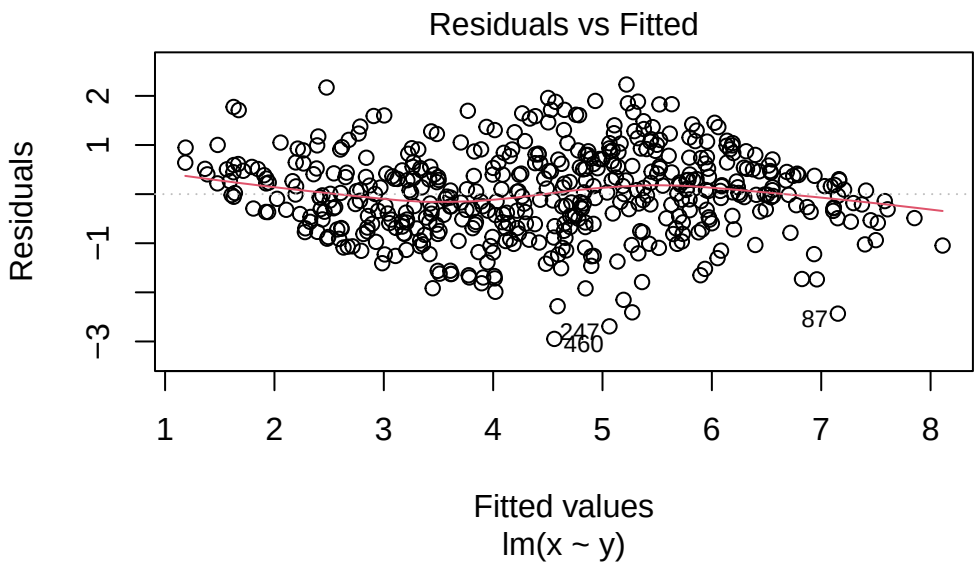


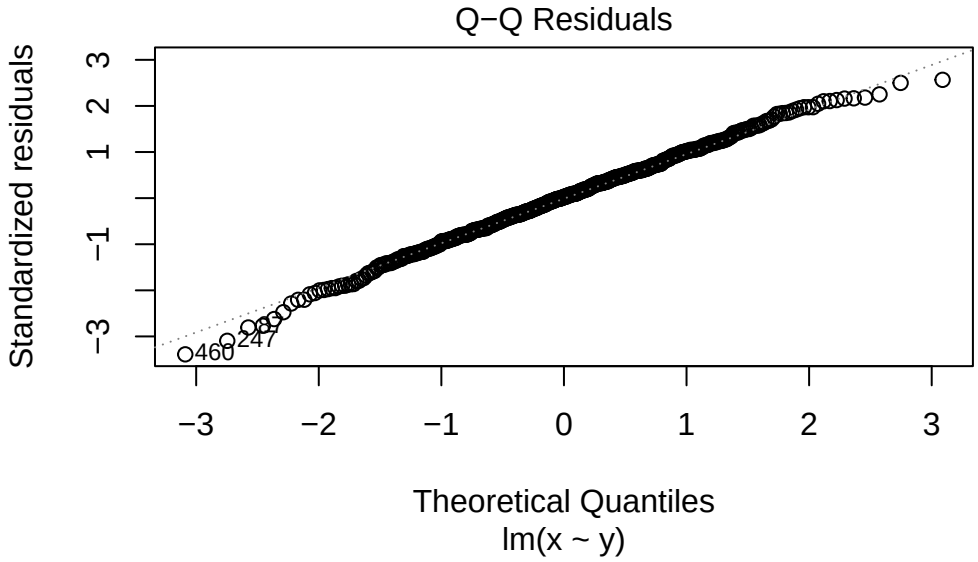

```r
model <- lm(x ~y)
summary(model)
```

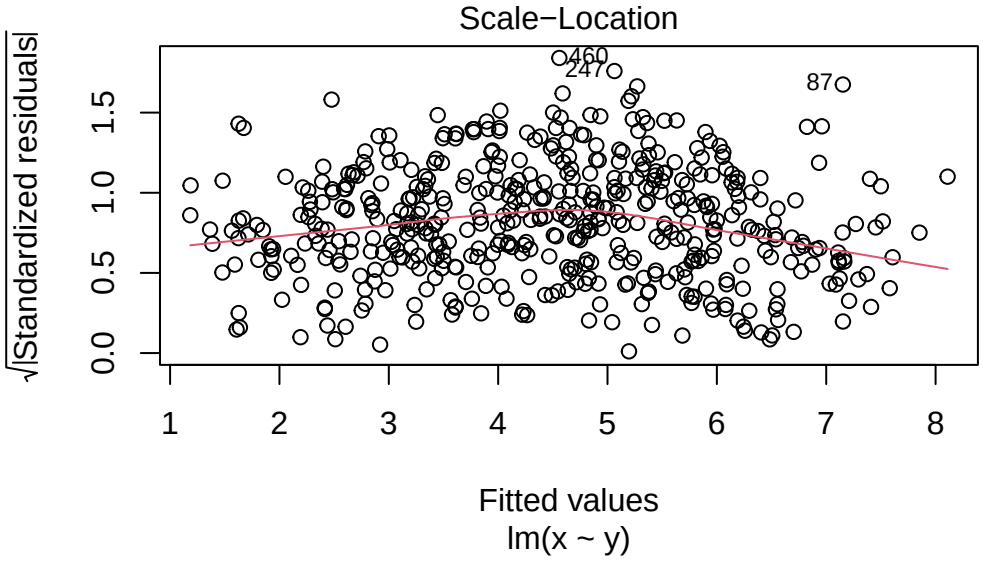
```
Call:
lm(formula = x ~ y)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.94796 -0.57730  0.02277  0.55661  2.23185 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.90533    0.10039   9.018   <2e-16 ***
y            1.04973    0.02726  38.505   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

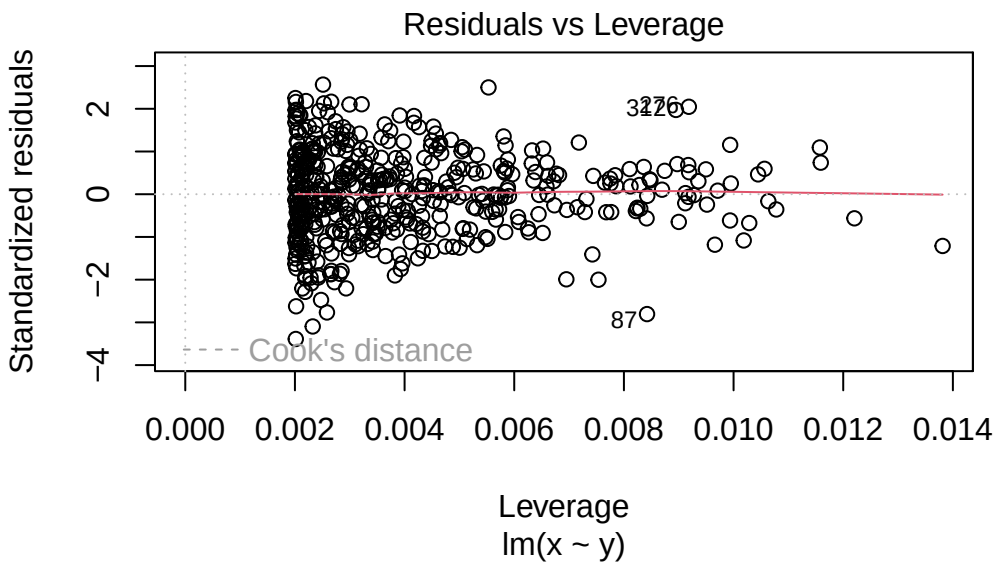
```
Residual standard error: 0.8708 on 496 degrees of freedom
Multiple R-squared:  0.7493,    Adjusted R-squared:  0.7488
F-statistic:  1483 on 1 and 496 DF,  p-value: < 2.2e-16
```

```r
plot(model)
```




Q−Q Residuals
Standardized residuals
3
2
1
−1
−3
460
247
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(x ~ y)


Scale−Location
√|Standardized residuals|
1.5
1.0
0.5
0.0
460
247
87
1
2
3
4
5
6
7
8
Fitted values
lm(x ~ y)




```r
# confidence interval for beta1
se <- summary(model)$coefficients[2, 2]  # standard error of beta1
beta1 <- summary(model)$coefficients[2, 1]  # estimate of beta1
n <- nrow(income_data)  # number of observations
t_value <- qt(0.975, df = n - 2)  # critical value for 95% confidence interval
lower_bound <- beta1 - t_value * se
upper_bound <- beta1 + t_value * se
cat("95% confidence interval for beta1: [", lower_bound, ", ", upper_bound, "]\n")
```

```
95% confidence interval for beta1: [ 0.9961633 ,  1.10329 ]
```

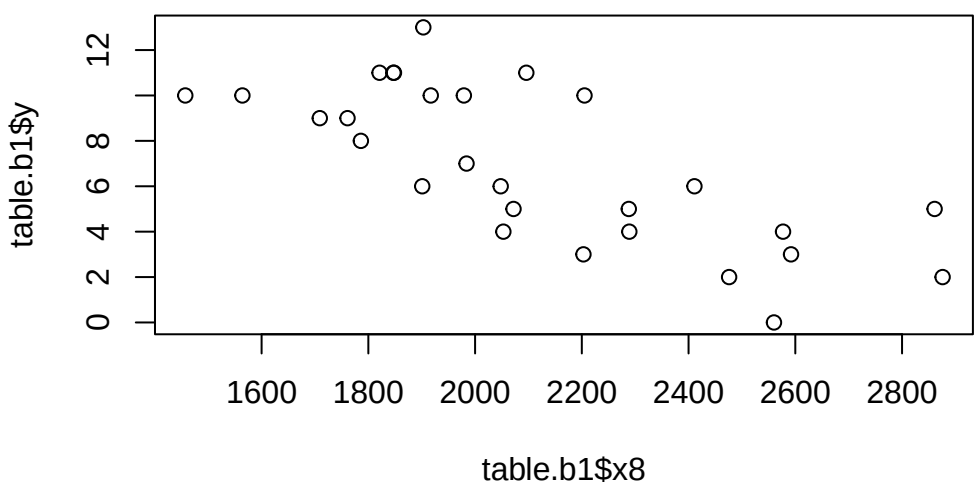
```r
library(MPV)
```

```
Loading required package: lattice

Warning: package 'lattice' was built under R version 4.5.2

Loading required package: KernSmooth

KernSmooth 2.23 loaded
Copyright M. P. Wand 1997-2009
```

```r
data(table.b1)
head(table.b1)
```

```
   y   x1   x2   x3   x4 x5  x6   x7   x8   x9
1 10 2113 1985 38.9 64.7  4 868 59.7 2205 1917
2 11 2003 2855 38.8 61.3  3 615 55.0 2096 1575
3 11 2957 1737 40.1 60.0 14 914 65.6 1847 2175
4 13 2285 2905 41.6 45.3 -4 957 61.4 1903 2476
5 10 2971 1666 39.2 53.8 15 836 66.1 1457 1866
6 11 2309 2927 39.7 74.1  8 786 61.0 1848 2339
```

```r
# attach(table.b1)
# detatch(table.b1)
plot(table.b1$x8, table.b1$y)
abline(model$coefficients[1],model$coefficients[2], col = "red", lty = "dashed")
```




```r
model <- lm(table.b1$y ~ table.b1$x8)
summary(model)
```

```
Call:
```

```
lm(formula = table.b1$y ~ table.b1$x8)

Residuals:
   Min     1Q Median     3Q    Max 
-3.804 -1.591 -0.647  2.032  4.580 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 21.788251   2.696233   8.081 1.46e-08 ***
table.b1$x8 -0.007025   0.001260  -5.577 7.38e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.393 on 26 degrees of freedom
Multiple R-squared:  0.5447,	Adjusted R-squared:  0.5272 
F-statistic:  31.1 on 1 and 26 DF,  p-value: 7.381e-06
```

```r
summary(aov(model))
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)    
table.b1$x8  1  178.1  178.09    31.1 7.38e-06 ***
Residuals   26  148.9    5.73                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# confidence interval for beta1
```

```r
library(MPV)
data(p8.3)
head(p8.3)
attach(p8.3)
par(mfrow = c(1,3))
plot(x1, x2)
plot(x1, y)
plot(x2, y)
x <- matrix(rbind(rep(1, length(y)),x1, x2), byrow = TRUE, ncol = 3)
# bind by row

beta_hat <- solve(t(x)%*%x)%*%t(x)%*%y
model <- lm(y ~ x1 + x2)
summary(model)
anova(model)
```